

MULTI-USER CALL BELL

For Up To Nine Persons

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This unique but simple call bell can be used to call an office boy or a help at home from up to nine different places. Besides ringing, it shows the number of caller's room or place from where the call is being made.

The author's prototype is shown in Fig. 1 while its block diagram is shown in Fig. 2. The call bell is built around LM7805 voltage regulator IC1, ICs 74LS147, 74LS04, and 74LS247 (IC2-IC4), NE555 timer IC5, BC547 transistors T1 and T2, and 1N4148 signal diodes D1-D4.

IC1 provides 5V regulated power supply to the circuit except piezo buzzer PZ1, which is connected to the 9V battery directly.

IC2 is 10-line to 4-line priority encoder. It encodes nine data lines to four-line BCD. Zero condition

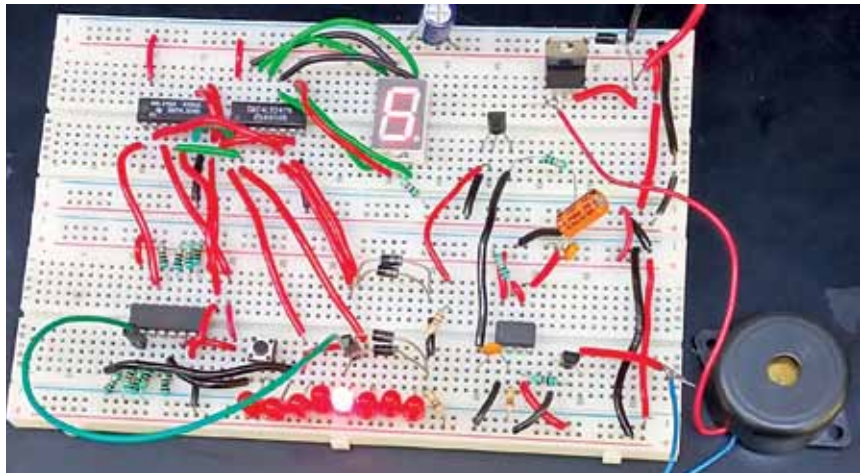


Fig. 1: Author's prototype

requires no input, as zero is encoded when all nine data lines are at a high logic level.

IC3 is hex inverter, which inverts output of 74LS147. Only gates IC3(A) through IC3(D) are used to invert the four-line BCD. The inverted outputs of IC3 are given to IC4 and the four diodes.

IC4 is a BCD-to-seven-segment decoder/driver. It has active-low outputs to directly drive the signal diodes. Output of IC4 is given to the common anode display LTS-6960HR/ LTS542 (DIS1), which is used to display the caller's room number.

The input switches (S1 through S9) are used to trigger timer IC5 through four signal diodes (D1 through D4). IC5 drives the piezo buzzer for a pre-set time.

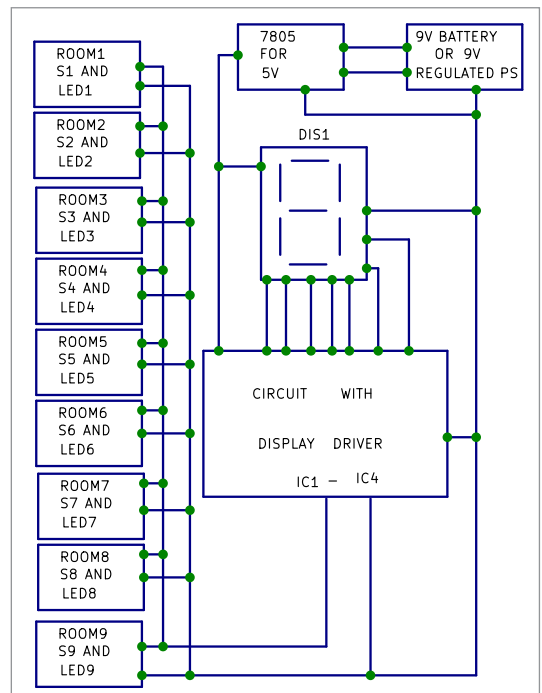


Fig. 2: Block diagram

Output of IC5 is decided by resistor R13 and capacitor C1.

The nine switches (S1 through

PARTS LIST

Semiconductors:

IC1	- LM7805, 5V voltage regulator
IC2	- 74LS147 10-line to 4-line priority encoder
IC3	- 74LS04 hex inverter
IC4	- 74LS247 7-segment decoder driver
IC5	- NE555 timer
T1, T2	- BC547 NPN transistor
D1-D4	- 1N4148 signal diode
LED1-LED9	- 5mm LED
DIS1	- 7-segment display LTS-6960HR/LTS542

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R9, R11, R14	- 1-kilo-ohm
R10, R12, R13,	
R16	- 10-kilo-ohm
R15	- 2.2-kilo-ohm

Capacitors:

C1	- 220 μ F, 35V electrolytic
C2	- 0.01 μ F ceramic disk
C3	- 0.1 μ F ceramic disk

Miscellaneous:

S1-S10	- On/off switch
PZ1	- Piezo buzzer
BATT.1	- 9V battery/regulated power supply